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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 02/16/2007 |
| **PI Name:** | Bellen, Hugo D.V.M., Ph.D. | | |
| **Project Title:** | The role of senseless in sensory organ precursor selection, and its vertebrate homologue, GFI1, in hair cell differentiation | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Developmental biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Animal Biology: Invertebrate | | |
| **Space Biology Cross-Element Discipline:** | (1) Developmental Biology<br>(2) Neurobiology | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 77030 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2003 Space Biology<br>03-OBPR-03 |
| **Start Date:** | 07/01/2004 | **End Date:** | 09/30/2006 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 3 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNA04CC46G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | Hair cells function as mechanoreceptors to transduce gravity, balance, and hearing. We have previously shown that the generation of hair cells is dependent on the expression of the Mouse atonal homologue 1 (MATH1)in precursors of hair cells. Most recently, injection of adenovirusus encoding Math1 into the inner ear fluid of adult animals has been shown to regenerate hair cells that are innervated. Unfortunately, regeneration is not very efficient, similar to what is observed when fly atonal alone is overexpressed in flies. However, coexpression of atonal and senseless, another transcription factor, is extremely efficient in specifying numerous extra sensory organs in flies. As we have previously shown that senseless, similar to atonal, is necessary and sufficient to form sensory organs in flies, we propose to study the role of senseless in much more detail in specific aim 1. In addition, we also propose to study the role of the vertebrate senseless homologue, Gfi1 in specific aim 2, as we have shown that this gene is also required for proper hair cell differentiation and function. We therefore wish to further our understanding of the biology of this gene in vivo, and test if the | | |

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| | synergism observed between atonal and senseless is evolutionary conserved. We hope that this research will allow us eventually to regenerate hair cells in adult animals that have lost functional hair cells. To understand how hair cells which are required for hearing and proprioception develop. To use this information to plan efficient strategies to regenerate hearing and propioception in humans that have lost these cells. The significance of our research has been highlighted recently with a publication in Nature Medicine showing that our premises that Math1 can regenerate hearing in deaf mammals is correct. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | We have made major progress in the past year on four different fronts. First, we have shown that senseless is a true proneural protein in certain lineages of the PNS. In the wing margin, senseless is required to specify the sensory organ precursors whereas proneural proteins of the bHLH family which normally are required to specify the SOP fates only play a role in suppressing apoptosis. In addition, this work also demonstrates that senseless plays very different roles in the development of the thoracic bristles, as loss of senseless in these cell lineages causes a transformation of the SOPIIA to an SOPIIb, as well as glial to neuron transformations. These data clearly show that the same gene which is expressed in similar cell lineages, can play quite different roles in the developmental hierarchy. This work was published in the May Development issue (Jafar-Nejad et al., 2006) Second, we have dissected the functional domains of senseless in vitro and in vivo and have shown that senseless acts as a transcriptional activator and repressor in a dosage dependent fashion. Our data indicate that senseless acts as transcriptional repressor when bound to DNA. It's central two Zn-fingers are specifically required for this function. When the concentration of senseless rises above a certain threshold it will bind to proneural proteins and strongly activate transcription. Senseless then strongly potentiates the transcriptional activity of proneural proteins. This work was just published on line in Development (Acar et al., 2006).<br>Third, in collaboration with Dr. Huda Zoghbi, we have shown that the mouse homologue of senseless, Gfi1, binds to the protein encoded by the Spinar Cerebellar Ataxia 1 gene in flies and mouse. Loss of Gfi1 in mice shows a progressive loss of the Purkinje Cerebellar neurons and ataxia. Our data indicate that overexpression of the SCA1 gene causes a proteasome induced degradation of Gfi1 in mice overexpressing the SCA1 protein. Loss of Gfi1 may then induce cell death of Purkinje Cells an Ataxia. This work was published in Cell (Tsuda et al., 2005).<br><br>Fourth, in collaboration with Dr. Y. Raphael at the University of Michigan, we have initiated studies to determine if Gfi1 synergizes with Math1 in the regeneration of hair cells. Based on our data in flies we proposed that Gfi1 and Math1 should synergize to produce hair cells in animals that were first treated with drugs that kill hair cells. We constructed a whole series of adenoviruses that are currently being injected in animals that lack hair cells and hearing. These experiments may allow us to regenerate hair cells in balance organs, thereby providing a much better proprioceptive input in patients that have lost hair cells. |
| **Bibliography Type:** | Description: (Last Updated: 09/03/2009) |
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